

RAG Glossary

Common terms and acronyms used in RAG systems.

A

ANN (Approximate Nearest Neighbor)

Algorithm that finds similar vectors quickly by trading perfect accuracy for speed. Essential for scaling vector search to millions of documents.

Augmentation

The process of adding retrieved context to a prompt before sending it to an LLM.

C

Chunk/Chunking

Splitting documents into smaller pieces for embedding and retrieval. Common sizes: 200-1000 characters.

ChromaDB

Open-source vector database designed for AI applications. Easy to use, runs locally.

Cosine Similarity

Metric for measuring how similar two vectors are. Range: -1 (opposite) to 1 (identical). Commonly used in RAG retrieval.

Context Window

Maximum number of tokens an LLM can process at once (input + output). GPT-3.5: 16K, GPT-4: 128K,

Claude 3: 200K.

Cross-Encoder

Model that scores query-document pairs directly for re-ranking. More accurate but slower than embeddings.

E

Embedding

Vector (list of numbers) representing text in high-dimensional space. Captures semantic meaning.

Embedding Model

Neural network that converts text to embeddings. Examples: text-embedding-ada-002, all-MiniLM-L6-v2.

F

Faiss

Facebook AI Similarity Search. High-performance library for vector search, especially at scale.

Few-Shot Prompting

Providing examples in the prompt to guide the LLM's response style.

Fine-Tuning

Training a model on specific data to specialize it. Alternative to RAG but more expensive to update.

G

Generation

The "G" in RAG. Using an LLM to create text based on retrieved context.

GPT

Generative Pre-trained Transformer. Family of LLMs by OpenAI (GPT-3.5, GPT-4).

H

Hallucination

When an LLM generates plausible-sounding but false information. RAG reduces this by grounding responses in retrieved facts.

HNSW (Hierarchical Navigable Small World)

Graph-based ANN algorithm. Very fast queries, high accuracy. Used by ChromaDB, Weaviate, Qdrant.

Hybrid Search

Combining semantic search (embeddings) with keyword search (BM25) for better results.

I

Indexing

Process of preparing documents for retrieval: chunking, embedding, and storing in vector database.

Inference

Using a trained model to make predictions (generate text, create embeddings, etc.).

IVF (Inverted File Index)

ANN algorithm that clusters vectors into regions for faster search. More memory efficient than HNSW.

K

Knowledge Base

Collection of documents/data that a RAG system retrieves from.

K-NN (K-Nearest Neighbors)

Finding the K most similar items. In RAG: finding K most relevant document chunks.

L

LLM (Large Language Model)

Neural network trained on massive text to understand and generate language. Examples: GPT-4, Claude, Llama.

M

Metadata

Information about documents (source, date, author, etc.). Used for filtering in RAG.

MMR (Maximum Marginal Relevance)

Retrieval strategy that balances relevance with diversity to avoid redundant results.

O

Ollama

Tool for running open-source LLMs locally. Useful for private/offline RAG systems.

P

Pinecone

Managed vector database service. Fully serverless, scales automatically.

Prompt Engineering

Art/science of crafting effective prompts to get desired LLM outputs.

Prompt Template

Reusable pattern for creating prompts. Example: "Context: {context}\nQuestion: {question}\nAnswer:"

Q

Query

User's question or search term in a RAG system.

Query Expansion

Generating variations of a query to retrieve more comprehensive results.

R

RAG (Retrieval-Augmented Generation)

Technique combining information retrieval with LLM generation for grounded, accurate responses.

Re-ranking

Re-ordering retrieved results using a more sophisticated model (often cross-encoder) for better relevance.

Retrieval

The "R" in RAG. Finding relevant documents/chunks from a knowledge base.

S

Semantic Search

Finding information based on meaning rather than exact keywords. Enabled by embeddings.

Sentence Transformers

Library and collection of models for creating embeddings. Popular for open-source RAG.

Similarity Score

Numerical measure of how related two pieces of text are. Higher = more similar.

T

Temperature

LLM parameter controlling randomness. 0 = deterministic, 1 = creative. For RAG, typically 0.3-0.7.

Token

Smallest unit of text for LLMs. Roughly 4 characters or 3/4 of a word. "Hello world" $\approx$ 2-3 tokens.

Top-K

Retrieving the K highest-scoring results. Common in RAG: K=3-5 chunks.

V

Vector

List of numbers representing data in mathematical space. Text embeddings are vectors.

Vector Database

Specialized database for storing and searching vectors efficiently. Examples: ChromaDB, Pinecone, Weaviate.

Vector Search

Finding similar vectors using distance/similarity metrics. Core of RAG retrieval.

Common Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ANN	Approximate Nearest Neighbor
API	Application Programming Interface
BM25	Best Matching 25 (keyword search algorithm)
DB	Database
FAISS	Facebook AI Similarity Search
GPT	Generative Pre-trained Transformer
HNSW	Hierarchical Navigable Small World
IVF	Inverted File Index
K-NN	K-Nearest Neighbors
LLM	Large Language Model
ML	Machine Learning

MMR	Maximum Marginal Relevance
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation

Quick Reference

Similarity Scores

- 1.0 = Identical
- 0.8-1.0 = Very similar
- 0.5-0.8 = Somewhat similar
- 0.0-0.5 = Not very similar
- -1.0 = Opposite (rare)

Typical RAG Parameters

- Chunk size: 500-1000 characters
- Chunk overlap: 50-200 characters
- Top-K retrieval: 3-5 chunks

- Temperature: 0.3-0.7
- Embedding dimensions: 384-1536

Model Comparison

Model Type	Speed	Quality	Cost
GPT-3.5	Fast	Good	Low
GPT-4	Slow	Excellent	High
Claude 3 Haiku	Fast	Good	Low
Claude 3 Opus	Slow	Excellent	High
Llama 3 (local)	Fast*	Good	Free**

Depends on hardware

*Hosting costs only

Need more help? Check the [lessons](#) for detailed explanations!